

2022.11.30
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ch6. 定积分的应用

§6.1 定积分的元方法.

定积分的定义再写一下

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \underbrace{f(\xi_i)}_{\text{相乘}} \underbrace{\Delta x_i}_{\text{求和}}, \text{取极限}$$

可加性
不一定是面积, 但可以参照面积来理解.

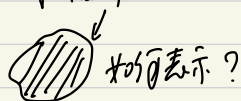
§6.2 定积分在几何上的应用.

点, 线, 面, 体
0, 1, 2, 3

① 求面积

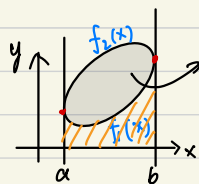
② 求体积

③ 求弧长



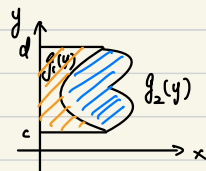
1. 直角坐标系.

用子出来给出边界



$$f_1(x) \leq y \leq f_2(x) \\ a \leq x \leq b$$

$$\text{面积} = \int_a^b (f_2(x) - f_1(x)) dx$$



$$g_1(y) \leq x \leq g_2(y) \\ c \leq y \leq d$$

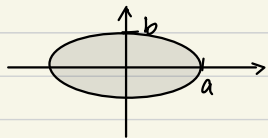
$$\text{面积} = \int_c^d (g_2(y) - g_1(y)) dy$$



可加性

例. 计算椭圆面积

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



$$1). -b\sqrt{1-\frac{x^2}{a^2}} \leq y \leq b\sqrt{1-\frac{x^2}{a^2}}, \quad -a \leq x \leq a$$

$$\text{面积 } S = \int_{-a}^a 2b\sqrt{1-\frac{x^2}{a^2}} dx \stackrel{\frac{x}{a}=t}{=} \pi ab$$

$$2). -a\sqrt{1-\frac{y^2}{b^2}} \leq x \leq a\sqrt{1-\frac{y^2}{b^2}}, \quad -b \leq y \leq b$$

$$\text{面积 } S = \int_{-b}^b 2a\sqrt{1-\frac{y^2}{b^2}} dy \stackrel{\frac{y}{b}=t}{=} \pi ab$$

2°. 极坐标公式



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

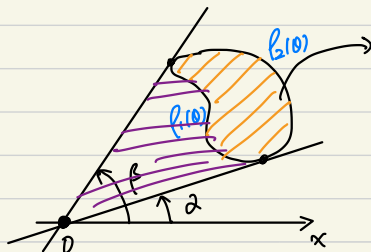
近似为扇形



$$S = \frac{1}{2} r^2 \theta$$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2} r_i^2 \Delta \theta_i$$

$$\int_{\alpha}^{\beta} \frac{1}{2} r^2(\theta) d\theta$$

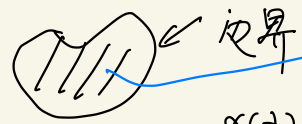


$$r_1(\theta) \leq r \leq r_2(\theta), \quad \alpha \leq \theta \leq \beta$$

~~$$\text{面积} \neq \int_{\alpha}^{\beta} (r_2(\theta) - r_1(\theta)) d\theta$$~~

$$\text{面积} = \frac{1}{2} \int_{\alpha}^{\beta} (r_2^2(\theta) - r_1^2(\theta)) d\theta$$

3. 参数方程.

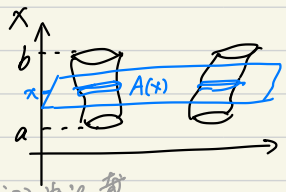


面积 = $\left| \int_a^b y(t) \underbrace{x'(t)}_{dx(t)} dt \right|$

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}, \quad a \leq t \leq b, \quad \begin{cases} x(a) = x(b) \\ y(a) = y(b) \end{cases}$$

② 体积.

祖冲之
祖暅定理
五世纪

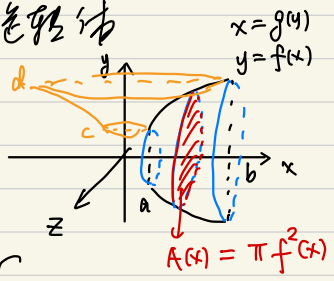


圆柱体体积 = $A(x) \Delta x$

1° $V = \int_a^b A(x) dx$

《九章算术》中记载

2° 旋转体

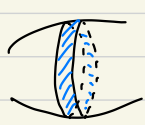


绕 x 轴: $V = \int_a^b \pi f^2(x) dx$

绕 y 轴: $V = \int_c^d \pi g^2(y) dy$

注: 画图看看.
不要直接套公式
例 8 (page 282)

补充: 侧面积 = $\int_a^b 2\pi f(x) ds(x) = \int_a^b 2\pi f(x) \sqrt{1+f'^2(x)} dx$



圆台

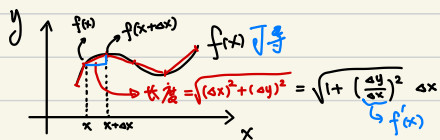


侧面积 $\approx ds \cdot 2\pi r$

参数: $S_{侧} = \int_a^b 2\pi y(t) \sqrt{x'^2(t) + y'^2(t)} dt$

更一般情形
见下册.

③ 弧长 = $\int_a^b \sqrt{1 + (f'(x))^2} dx$



→ 丁号

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}, \alpha \leq t \leq \beta$$

$$\text{弧长} = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

$$\sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2} \Delta t$$

弧长微元

$$ds = \sqrt{x'(t)^2 + y'(t)^2} dt$$

$$ds = \sqrt{1 + f'(x)^2} dx$$

$$\text{弧长} = \int_a^b ds$$

例: 参数曲线. 附录 III. page 371-373, it 弧长, 面积.

例: $\rho(\theta) = a\theta$, $\theta \in [0, 2\pi]$

阿基米德螺旋线



$$\begin{cases} x(\theta) = \rho(\theta) \cos \theta \\ y(\theta) = \rho(\theta) \sin \theta \end{cases}$$

$$\text{面积} = \int_0^{2\pi} \frac{1}{2} \rho^2(\theta) d\theta = \int_0^{2\pi} \frac{a^2}{2} \theta^2 d\theta = \frac{a^2}{2} \frac{(2\pi)^3}{3} = \frac{4}{3} a^2 \pi^3$$

$$\text{弧长} = \int_0^{2\pi} ds = \int_0^{2\pi} \sqrt{x'(\theta)^2 + y'(\theta)^2} d\theta = \int_0^{2\pi} \sqrt{a^2 \theta^2 + a^2} d\theta = a^2 \int_0^{2\pi} \sqrt{1 + \theta^2} d\theta$$

page 286 例 13

例: $\rho(\theta) = a(1 + \cos \theta)$, $\theta \in [0, 2\pi]$

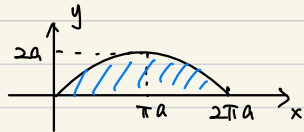


心型线

$$\text{面积} = \int_0^{2\pi} \frac{1}{2} \rho^2(\theta) d\theta = \int_0^{2\pi} \frac{1}{2} a^2 (1 + \cos \theta)^2 d\theta = \frac{3}{2} \pi a^2 \quad (\text{例 5})$$

$$\text{弧长} = \int_0^{2\pi} ds = \int_0^{2\pi} \sqrt{a^2 (1 + \cos \theta)^2 + a^2 \sin^2 \theta} d\theta = 8a$$

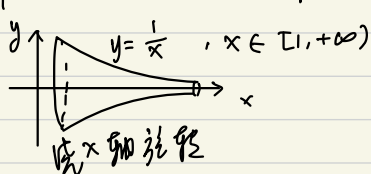
例: $\begin{cases} x(t) = a(t - \sin t) \\ y(t) = a(1 - \cos t) \end{cases}$, $t \in [0, 2\pi]$ 摆线



面积: $\int_0^{2\pi} y(t) x'(t) dt = 3\pi a^2$

弧长: $\int_0^{2\pi} \sqrt{x'(t)^2 + y'(t)^2} dt = 8a$ (例12)

例: Torricelli's Trumpet (托里拆利小号), 体积有限, 表面积无限.

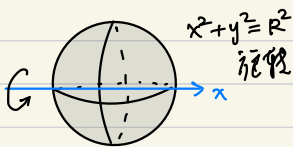


$f(x) = \frac{1}{x^p}$, $p > 0$
体积: $\int_1^{+\infty} \pi f^2(x) dx = \int_1^{+\infty} \pi \frac{dx}{x^{2p}} < +\infty$ 若 $2p > 1$

表面积: $\int_1^{+\infty} 2\pi f(x) ds = \int_1^{+\infty} \frac{2\pi}{x^p} \sqrt{1 + \frac{p^2}{x^{2p+2}}} dx = +\infty$ $0 < p \leq 1$

也利: Gabriel's Horn, $V = \int_1^{+\infty} \frac{2}{x^2} dx = \pi$, $S = +\infty$

例: 球的体积和表面积.



$V = \int_{-R}^R \pi f^2(x) dx = \int_{-R}^R \pi (R^2 - x^2) dx = 2\pi R^3 - \frac{2}{3}\pi R^3 = \frac{4}{3}\pi R^3$

表面积: $S = \int_{-R}^R 2\pi f(x) ds = \int_{-R}^R 2\pi \sqrt{R^2 - x^2} \sqrt{1 + \left(\frac{-x}{R^2 - x^2}\right)^2} dx = \int_{-R}^R 2\pi R dx = 4\pi R^2$

注: 祖暅利用牟合方盖得到了球体积的计算公式. (下册再详细讲)

注: 椭圆周长, 椭圆球的表面积如何算? 答: 没有简单的表达式!

2022.12.2

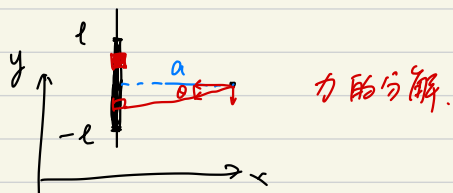
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§6.3. 定积分在物理学上的应用. (不考)

可加性

做功 $W = F \cdot s$, 压力 $P = p \cdot A$, 引力 $F = G \frac{m_1 m_2}{r^2}$

sol. page 293. sol 5

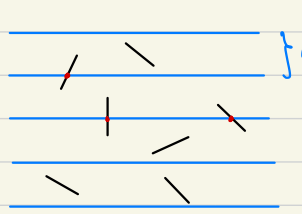


$$F = - \int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{G m M}{(\sqrt{a^2 + y^2})^3} dy \quad \leftarrow \quad - \int_{\frac{l}{2}}^{\frac{l}{2}} \frac{G m M}{a^2 + y^2} \underbrace{\cos \theta}_{\frac{a}{\sqrt{a^2 + y^2}}} dy$$

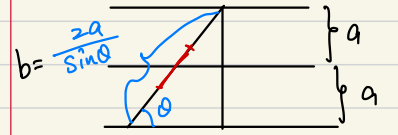
补元 § 6.4. 定积分在其他方面的应用. (不考) eg. 概率论

eg. 正态分布 $\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} dx = 1$

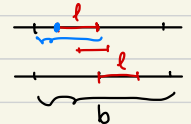
eg. Buffon's needle problem 蒲丰投针问题



针与线相交 概率 $= \frac{2l}{\pi a}$
若 $a=l$ $\rightarrow \frac{2}{\pi}$



$$p = \int_0^{2\pi} \frac{1}{2\pi} \underbrace{\frac{l}{|a/\sin\theta|}}_{\downarrow} d\theta = \int_0^{2\pi} \frac{l}{2\pi a} |\sin\theta| d\theta = \frac{4l}{2\pi a} = \frac{2l}{\pi a}$$

注:  相交, 得是这种情形. 关注左端点, 可知概率为 $\frac{l}{b/2} = \frac{l}{|a/\sin\theta|}$

注: 若 $l > a$, 则 $p = \int_0^{2\pi} \frac{1}{2\pi} \min(1, \frac{l}{a} |\sin\theta|) d\theta$, 结果稍复杂一些.